

Hydrodynamic limits of the Chern–Simons gauged models

Shih-Hsien Fest on Hyperbolic Conservation Laws and Kinetic Theory

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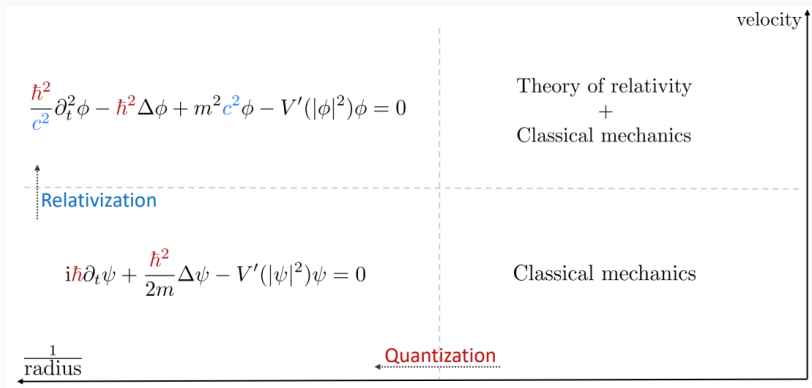
1 Introduction

- Non-relativistic and Semi-classical limits
- Chern-Simons gauged models

2 Hydrodynamic limit of the Chern–Simons–Higgs equations

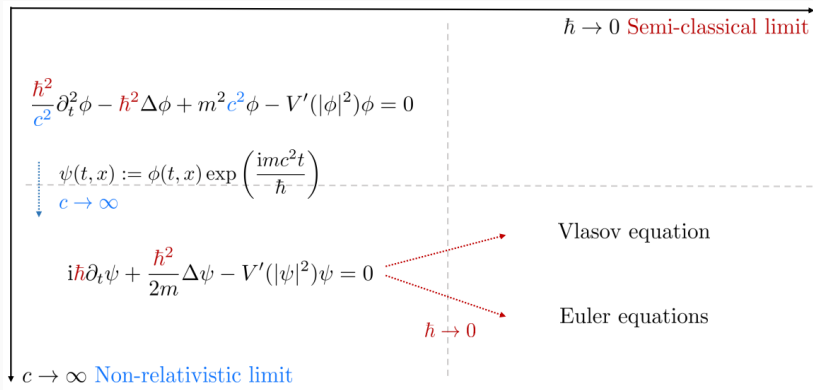
3 Remarks

Non-relativistic and Semi-classical limits



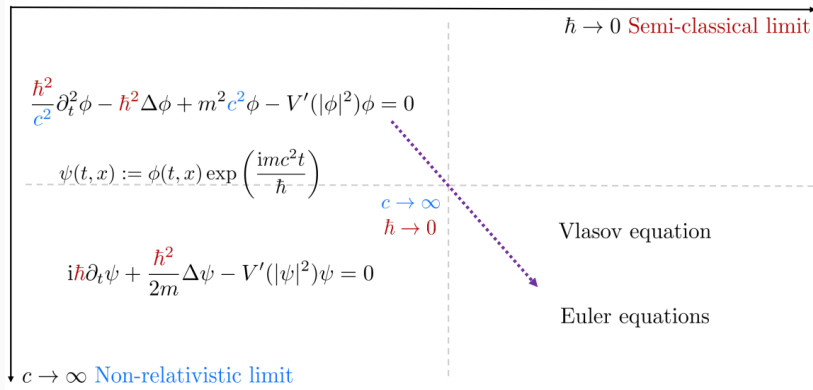
- c : the speed of light = 299,792,458 m/s
- $\hbar$: the (reduced) Planck constant $\approx 1.054571817 \times 10^{-34} \text{ J} \cdot \text{s}$

Non-relativistic and Semi-classical limits



- c : the speed of light
- $\hbar$: the (reduced) Planck constant

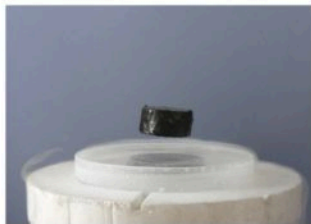
Non-relativistic and Semi-classical limits



- c : the speed of light
- $\hbar$: the (reduced) Planck constant

The Chern-Simons gauge theory

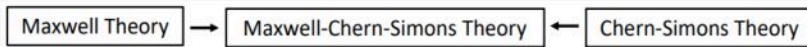
was proposed to explain the planar electromagnetic phenomena:



High-Temperature Superconductivity

- Low- T_c superconductivity
(1911, H. K. Onnes) Mercury $T_c = 4.2K$
(1950, V. L. Ginzburg-K.D.Landau) **Maxwell**-Higgs
(1957) BCS Theory
- High- T_c superconductivity
(1986, K.A. Müller-J.G.Bednorz) $T_c = 35K - 138K$
(1990) Chern-Simons Theory
Chern-Simons-Higgs/ Dirac/ Schrödinger

- To describes the evolution of the planar quantum particles, interacting with the electromagnetic fields generated by itself:



The Chern–Simons–Higgs model

- Lagrangian density of Chern–Simons–Higgs model in $\mathbb{R}^{2+1}$:

$$\mathcal{L}^R := \frac{\kappa}{2c} \epsilon^{\rho\mu\nu} A_\rho F_{\mu\nu} + \hbar^2 D_\mu \phi \overline{D^\mu \phi} - m^2 c^2 |\phi|^2 - V(|\phi|^2)$$

- κ : Chern-Simons coupling constant
- $\epsilon^{\rho\mu\nu}$: totally antisymmetric tensor with $\epsilon^{012} = 1$
- $F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu$: electromagnetic field
- $D_\mu := \partial_\mu - \frac{i}{c\hbar} A_\mu$: covariant derivative with $\partial_0 = \frac{1}{c} \partial_t$, $\partial_j = \partial_{x_j}$
- $\phi : \mathbb{R}^{2+1} \rightarrow \mathbb{C}$: matter field, $A_\mu : \mathbb{R}^{2+1} \rightarrow \mathbb{R}$: gauges ($\mu = 0, 1, 2$)

The Chern–Simons–Higgs equations

- Chern–Simons–Higgs(CSH) equations:

$$\hbar^2 D_\mu D^\mu \phi + m^2 c^2 \phi + V'(|\phi|^2)\phi = 0, \quad t > 0, \quad x \in \mathbb{R}^2,$$

$$\kappa \left(\frac{1}{c} \partial_t A_1 - \partial_1 A_0 \right) = -2\hbar \operatorname{Im}(\bar{\phi} D_2 \phi), \quad \kappa \left(\frac{1}{c} \partial_t A_2 - \partial_2 A_0 \right) = 2\hbar \operatorname{Im}(\bar{\phi} D_1 \phi),$$

$$\frac{\kappa}{c} (\partial_1 A_2 - \partial_2 A_1) = \frac{2\hbar}{c} \operatorname{Im}(\bar{\phi} D_0 \phi)$$

- $\phi : \mathbb{R}^{2+1} \rightarrow \mathbb{C}$: matter field,
- $A_\mu : \mathbb{R}^{2+1} \rightarrow \mathbb{R}$: gauge fields ($\mu = 0, 1, 2$)
- $D_\mu := \partial_\mu - \frac{i}{c\hbar} A_\mu$: covariant derivative with $\partial_0 = \frac{1}{c} \partial_t$, $\partial_j = \partial_{x_j}$
- c : speed of light, $\hbar$: Planck constant, m : mass
- V : self-interacting potential energy density ($V(\rho) = \frac{1}{\gamma-1} \rho^\gamma$, $\gamma > 1$)

The modulated CSH equations

- Modulated wave function:

$$\psi(t, x) := \sqrt{2m}\phi(t, x) \exp\left(\frac{imc^2 t}{\hbar}\right)$$

- Modulated CSH equations:

$$i\hbar D_t \psi - \frac{\hbar^2}{2m\mathbf{c}^2} D_t D_t \psi + \frac{\hbar^2}{2m} D_j D_j \psi - \frac{1}{2m} V' \left(\frac{1}{2m} |\psi|^2 \right) \psi = 0,$$

$$\kappa \left(\frac{1}{\mathbf{c}} \partial_t A_1 - \partial_1 A_0 \right) = -\frac{\hbar}{m} \operatorname{Im}(\bar{\psi} D_2 \psi), \quad \kappa \left(\frac{1}{\mathbf{c}} \partial_t A_2 - \partial_2 A_0 \right) = \frac{\hbar}{m} \operatorname{Im}(\bar{\psi} D_1 \psi),$$

$$\frac{\kappa}{\mathbf{c}} (\partial_1 A_2 - \partial_2 A_1) = -|\psi|^2 + \frac{\hbar}{m\mathbf{c}^2} \operatorname{Im}(\bar{\psi} D_t \psi),$$

$$\bullet \quad D_t := \mathbf{c} D_0 = \partial_t - \frac{i}{\hbar} A_0, \quad D_j = \partial_{x_j} - \frac{i}{\mathbf{c}\hbar} A_j$$

Remark: Non-relativistic limit

- In the limit of $c \rightarrow \infty$, the modulated CSH equations approach the following Chern–Simons–Schrödinger(CSS) equations:

$$i\hbar D_t \psi + \frac{\hbar^2}{2m} (D_1 D_1 \psi + D_2 D_2 \psi) - \tilde{V}'(|\psi|^2) \psi = 0,$$

$$\kappa \left(\partial_t \left(\frac{1}{c} A_1 \right) - \partial_1 A_0 \right) = -\frac{\hbar}{m} \text{Im}(\bar{\psi} D_2 \psi), \quad \kappa \left(\partial_t \left(\frac{1}{c} A_2 \right) - \partial_2 A_0 \right) = \frac{\hbar}{m} \text{Im}(\bar{\psi} D_1 \psi),$$

$$\kappa \left(\partial_1 \left(\frac{1}{c} A_2 \right) - \partial_2 \left(\frac{1}{c} A_1 \right) \right) = -|\psi|^2$$

$$\bullet \quad D_t = \partial_t - \frac{i}{\hbar} A_0, \quad D_j = \partial_{x_j} - \frac{i}{\hbar} \left(\frac{1}{c} A_j \right)$$

- Formal derivation, [1990, R. Jackiw and S.-Y. Pi]
- Rigorous derivation, [2009, M. Chae and H. Huh], [preprint, H. Huh and M.]

One parameterized CSH equations for hydrodynamic limit

- Define the scaling parameter $\varepsilon := \hbar$ and $\varepsilon^\delta := c^{-1}$, with $\delta > 0$.
- ε -modulated CSH equations: $(\psi^\varepsilon, A_0^\varepsilon, \varepsilon^\delta A^\varepsilon)$ satisfies

$$i\varepsilon D_t^\varepsilon \psi^\varepsilon - \frac{\varepsilon^{2+2\delta}}{2} D_t^\varepsilon D_t^\varepsilon \psi^\varepsilon + \frac{\varepsilon^2}{2} D_j^\varepsilon D_j^\varepsilon \psi^\varepsilon - \frac{1}{2} V' \left(\frac{1}{2} |\psi^\varepsilon|^2 \right) \psi^\varepsilon = 0,$$

$$\partial_t (\varepsilon^\delta A^\varepsilon) - \nabla A_0^\varepsilon = \varepsilon \operatorname{Im}(\overline{\psi^\varepsilon} D^\varepsilon \psi^\varepsilon)^\perp,$$

$$\nabla \times (\varepsilon^\delta A^\varepsilon) = -|\psi^\varepsilon|^2 + \varepsilon^{1+2\delta} \operatorname{Im}(\overline{\psi^\varepsilon} D_t^\varepsilon \psi^\varepsilon).$$

$$\cdot \quad D_t^\varepsilon = \partial_\mu - \frac{i}{\varepsilon} A_0^\varepsilon, \quad D_j^\varepsilon = \partial_{x_j} - \frac{i}{\varepsilon} (\varepsilon^\delta A_j^\varepsilon)$$

$$\cdot \quad D^\varepsilon := \begin{pmatrix} D_1^\varepsilon \\ D_2^\varepsilon \end{pmatrix} = \nabla - \frac{i}{\varepsilon} \varepsilon^\delta A^\varepsilon, \quad A^\varepsilon := \begin{pmatrix} A_1^\varepsilon \\ A_2^\varepsilon \end{pmatrix}$$

- Madelung Transformation:

$$\psi^\varepsilon(x, t) = \sqrt{\rho^\varepsilon(t, x)} \exp\left(\frac{i}{\varepsilon} S^\varepsilon(t, x)\right)$$

$$\bullet \rho^\varepsilon, S^\varepsilon : \mathbb{R}^{2+1} \rightarrow \mathbb{R}$$

- Hydrodynamic variables:

$$\rho^\varepsilon := |\psi^\varepsilon|^2, \quad J^\varepsilon := \rho^\varepsilon u^\varepsilon := \rho^\varepsilon (\nabla S^\varepsilon - \varepsilon^\delta A^\varepsilon) = \varepsilon \operatorname{Im}(\overline{\psi^\varepsilon} D^\varepsilon \psi^\varepsilon),$$

$$\rho_R^\varepsilon := \varepsilon^{1+2\delta} \operatorname{Im}(\overline{\psi_0^\varepsilon} D_t^\varepsilon \psi^\varepsilon), \quad J_R^\varepsilon := \varepsilon^{2+2\delta} \operatorname{Re}(\overline{D_t^\varepsilon \psi^\varepsilon} D^\varepsilon \psi^\varepsilon)$$

Formal Hydrodynamic limit

- Quantum Euler equations(with relativistic terms) with CS gauge fields:

$$\partial_t(\rho^\varepsilon - \rho_R^\varepsilon) + \nabla \cdot (\rho^\varepsilon u^\varepsilon) = 0,$$

$$\partial_t((\rho^\varepsilon - \rho_R^\varepsilon)u^\varepsilon) + \nabla \cdot (\rho^\varepsilon u^\varepsilon \otimes u^\varepsilon) + \nabla p(\rho^\varepsilon) = \frac{\varepsilon^2 \rho^\varepsilon}{2} \nabla \left(\frac{\square_\delta \sqrt{\rho^\varepsilon}}{\sqrt{\rho^\varepsilon}} \right),$$

$$\partial_t(\varepsilon^\delta A^\varepsilon) - \nabla A_0^\varepsilon = (\rho^\varepsilon u^\varepsilon)^\perp, \quad \nabla \times (\varepsilon^\delta A^\varepsilon) = -\rho^\varepsilon + \rho_R^\varepsilon$$

- Euler equations with CS gauge fields:

$$\partial_t \rho + \nabla \cdot (\rho u) = 0,$$

$$\partial_t(\rho u) + \nabla \cdot (\rho u \otimes u) + \nabla p(\rho^\varepsilon) = 0, \quad (\text{ECS})$$

$$\partial_t A - \nabla A_0 = (\rho u)^\perp, \quad \nabla \times A = -\rho$$

- $p(\rho) = (\gamma - 1)V\left(\frac{1}{2}\rho\right) = 2^{-\gamma}\rho^\gamma$: isentropic type pressure law
- $\square_\delta := -\varepsilon^{2\delta}\partial_t^2 + \Delta$

Property: Conservation Laws

(1) (Charge):

$$\partial_t (\rho^\varepsilon - \rho_R^\varepsilon) + \nabla \cdot J^\varepsilon = 0.$$

(2) (Current):

$$\begin{aligned} \partial_t (J^\varepsilon - J_R^\varepsilon) + \frac{\varepsilon^2}{2} \nabla \cdot (D^\varepsilon \psi^\varepsilon \otimes \overline{D^\varepsilon \psi^\varepsilon} + \overline{D^\varepsilon \psi^\varepsilon} \otimes D^\varepsilon \psi^\varepsilon) \\ + \nabla p(|\psi^\varepsilon|^2) = -\frac{\varepsilon^2}{2} (\varepsilon^{2\delta} \partial_t^2 - \Delta) \operatorname{Re} (\overline{\psi^\varepsilon} D^\varepsilon \psi^\varepsilon) \end{aligned}$$

(3) (Energy):

$$\mathcal{E}^\varepsilon(t) := \int_{\mathbb{R}^2} \left(\frac{1}{2} |\varepsilon D^\varepsilon \psi^\varepsilon|^2 + \frac{1}{2} |\varepsilon^{1+\delta} D_t^\varepsilon \psi^\varepsilon|^2 + V \left(\frac{1}{2} |\psi^\varepsilon|^2 \right) \right) (x, t) dx = \mathcal{E}(0)$$

Property: Gauge Invariance

- The CSH system is invariant under the gauge transform:

$$\phi \rightarrow \phi e^{-i\chi}, \quad A_\mu \rightarrow A_\mu - c\hbar \partial_\mu \chi, \quad \chi : \mathbb{R}^{2+1} \rightarrow \mathbb{R} \text{ smooth function.}$$

$$\Rightarrow \text{Coulomb gauge condition: } \nabla \cdot A = \partial_1 A_1 + \partial_2 A_2 = 0$$

- Equations for the gauge fields:

$$\Delta A_0^\varepsilon = 2\varepsilon \nabla \cdot \text{Im}(\overline{\phi^\varepsilon} D^\varepsilon \phi^\varepsilon)^\perp, \quad \Delta(\varepsilon^\delta A^\varepsilon) = 2\varepsilon^{1+2\delta} \nabla^\perp \text{Im}(\overline{\phi^\varepsilon} D_t^\varepsilon \phi^\varepsilon)$$

Assumptions on Initial Data

- Coulomb gauge condition:

$$\nabla \cdot A_{\text{in}}^\varepsilon = 0, \quad \nabla \times A_{\text{in}}^\varepsilon = -|\phi_{\text{in}}^\varepsilon|^2$$

Results on global well-posedness of the CSH equation:

- [(2002), D.Chae and K.Choe] For $\phi_{\text{in}} \in H^2, \partial_t \phi_{\text{in}}, A_{\text{in}} \in H^1$
- [(2007) H. Huh] For $\phi_{\text{in}} \in H^{1+\epsilon}, \partial_t \phi_{\text{in}}, A_{\text{in}} \in H^\epsilon, \epsilon > 0$

- Well-prepared initial data condition:

$$\mathcal{H}^\varepsilon(0) + \|\varepsilon^\delta A_{\text{in}}^\varepsilon - A_{\text{in}}\|_{L^{\frac{2\gamma}{\gamma+1}}} \leq C\varepsilon^\lambda$$

- Modulated energy:

$$\mathcal{H}^\varepsilon(t) = \int_{\mathbb{R}^2} \frac{1}{2} |(\varepsilon D^\varepsilon - iu)\psi^\varepsilon|^2 + \frac{1}{2} |\varepsilon^{1+\delta} D_t^\varepsilon \psi^\varepsilon|^2 + \frac{1}{\gamma-1} p(|\psi^\varepsilon|^2|\rho) \, dx$$

where $p(n|\rho) := 2^{-\gamma} (n^\gamma - \rho^\gamma - \gamma\rho^{\gamma-1}(n - \rho))$: relative pressure.

Let $(\psi^\varepsilon, A_\mu^\varepsilon)$ be the global solution to the modulated CSH system, and (ρ, u, A_μ) be the local smooth solution to the ECS up to time T_* .

(1) If $\gamma \geq 2$, we have the following convergence: as $\varepsilon \rightarrow 0$,

$$\rho^\varepsilon \rightarrow \rho \quad \text{in} \quad L^\infty([0, T_*]; L^\gamma(\mathbb{R}^2)),$$

$$J^\varepsilon \rightarrow \rho u \quad \text{in} \quad L^\infty([0, T_*]; L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)),$$

$$\rho_R^\varepsilon \rightarrow 0 \quad \text{in} \quad L^\infty([0, T_*]; L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)),$$

$$J_R^\varepsilon \rightarrow 0 \quad \text{in} \quad L^\infty([0, T_*]; L^1(\mathbb{R}^2)),$$

$$A_0^\varepsilon \rightarrow A_0 \quad \text{in} \quad L^\infty([0, T_*]; L^{2\gamma}(\mathbb{R}^2)),$$

$$\nabla A_0^\varepsilon \rightarrow \nabla A_0 \quad \text{in} \quad L^\infty([0, T_*]; L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)),$$

$$\varepsilon^\delta A^\varepsilon \rightarrow A \quad \text{in} \quad L^\infty([0, T_*]; L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)).$$

Theorem [Kim-M., Preprint]

Let $(\psi^\varepsilon, A_\mu^\varepsilon)$ be the global solution to the modulated CSH system, and (ρ, u, A_μ) be the local smooth solution to the ECS up to time T_* .

(2) If $1 < \gamma < 2$, we have the following convergence: as $\varepsilon \rightarrow 0$,

$$\rho^\varepsilon \rightarrow \rho \quad \text{in} \quad L^\infty([0, T_*]; L_{\text{loc}}^\gamma(\mathbb{R}^2)),$$

$$J^\varepsilon \rightarrow \rho u \quad \text{in} \quad L^\infty([0, T_*]; L_{\text{loc}}^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)),$$

$$\rho_R^\varepsilon \rightarrow 0 \quad \text{in} \quad L^\infty([0, T_*]; L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)),$$

$$J_R^\varepsilon \rightarrow 0 \quad \text{in} \quad L^\infty([0, T_*]; L^1(\mathbb{R}^2)),$$

$$A_0^\varepsilon \rightarrow A_0 \quad \text{in} \quad L^\infty([0, T_*]; L_{\text{loc}}^{2\gamma}(\mathbb{R}^2)),$$

$$\nabla A_0^\varepsilon \rightarrow \nabla A_0 \quad \text{in} \quad L^\infty([0, T_*]; L_{\text{loc}}^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)),$$

$$\varepsilon^\delta A^\varepsilon \rightarrow A \quad \text{in} \quad L^\infty([0, T_*]; L_{\text{loc}}^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)).$$

Proof of Theorem: Step 1. Modulated Energy Estimates

- The Modulated Energy:

$$\begin{aligned}\mathcal{H}^\varepsilon(t) &= \int_{\mathbb{R}^2} \frac{1}{2} |(\varepsilon D^\varepsilon - iu)\psi^\varepsilon|^2 + \frac{1}{2} |\varepsilon^{1+\delta} D_t^\varepsilon \psi^\varepsilon|^2 + \frac{1}{\gamma-1} p(\rho^\varepsilon|\rho) \, dx \\ &= \mathcal{E}^\varepsilon(t) - \int_{\mathbb{R}^2} J^\varepsilon \cdot u + \frac{1}{2} \rho^\varepsilon |u|^2 + \frac{1}{2\gamma} \left(\rho - \frac{\gamma}{\gamma-1} \rho^\varepsilon \right) \rho^{\gamma-1} \, dx\end{aligned}$$

Proposition. Under the same assumption on Theorem, for $0 \leq t \leq T_*$, we have

$$\begin{aligned}\mathcal{H}^\varepsilon(t) &\leq C\varepsilon^\alpha, \quad \text{where } \alpha = \min\{1, \delta, \lambda\} \\ &(\rightarrow 0 \quad \text{as } \varepsilon \rightarrow 0)\end{aligned}$$

Proof of Theorem: Step 2. Convergence Estimates

- The Modulated Energy:

$$\begin{aligned}\mathcal{H}^\varepsilon(t) &= \int_{\mathbb{R}^2} \frac{1}{2} |(\varepsilon D^\varepsilon - iu)\psi^\varepsilon|^2 + \frac{1}{2} |\varepsilon^{1+\delta} D_t^\varepsilon \psi^\varepsilon|^2 + \frac{1}{\gamma-1} p(\rho^\varepsilon|\rho) \, dx \\ &= \int_{\mathbb{R}^2} \frac{1}{2} \rho^\varepsilon |u^\varepsilon - u|^2 + \frac{1}{2} |\varepsilon \nabla \sqrt{\rho^\varepsilon}|^2 + \frac{1}{2} |\varepsilon^{1+\delta} D_t^\varepsilon \psi^\varepsilon|^2 + \frac{1}{\gamma-1} p(\rho^\varepsilon|\rho) \, dx\end{aligned}$$

- (Convergence of ρ^ε , J^ε , ρ_R^ε , J_R^ε):

$$\|\rho^\varepsilon - \rho\|_{L^\gamma} \leq C\varepsilon^{\frac{\alpha}{\gamma}}, \quad \|J^\varepsilon - \rho u\|_{L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)} \leq C\varepsilon^{\frac{\alpha}{\gamma}},$$

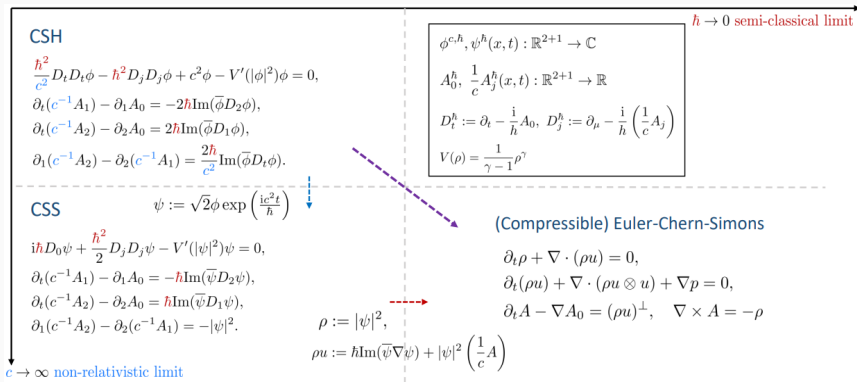
$$\|\rho_R^\varepsilon\|_{L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)} \leq C\varepsilon^\delta, \quad \|J_R^\varepsilon\|_{L^1(\mathbb{R}^2)} \leq C\varepsilon^\delta$$

- (Convergence of A_0^ε and A^ε):

$$\|A_0^\varepsilon - A_0\|_{L^{2\gamma}(\mathbb{R}^2)} + \|\nabla(A_0^\varepsilon - A_0)\|_{L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)} \leq \|\rho^\varepsilon u^\varepsilon - \rho u\|_{L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)} \leq C\varepsilon^{\frac{\alpha}{\gamma}},$$

$$\|\varepsilon^\delta A^\varepsilon - A\|_{L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)} \leq \|\varepsilon^\delta A_{\text{in}}^\varepsilon - A_{\text{in}}\|_{L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)} + CT^* \|\rho^\varepsilon u^\varepsilon - \rho u\|_{L^{\frac{2\gamma}{\gamma+1}}(\mathbb{R}^2)}$$

Conclusion



References:

- Formal limit [1990, R. Jackiw and S.-Y. Pi]
- Non-relativistic limit [2009, M. Chae and H. Huh] [preprint, H. Huh and M.]
- Semi-classical limit [2022, J. Kim and M.]
- Hydrodynamic limit of the Klein-Gordon equation [2012, C.-K. Lin and K.-C. Wu]

THANK YOU FOR YOUR ATTENTION!